

Natural Language and Speech Processing

Introduction

This report documents a sequence of laboratory exercises conducted over four weeks, covering fundamental concepts in Natural Language Processing (NLP) and Speech Processing. The progression moves from classical text preprocessing techniques to deep learning models for text and speech understanding.

Week 1: Text Preprocessing and Classical NLP

The first week focused on core NLP preprocessing techniques. Tokenization and stemming were implemented using the NLTK library, with stopwords removal to reduce noise in textual data. spaCy was explored as an alternative for efficient tokenization and lemmatization.

A TF-IDF based sentiment classifier was built using the IMDb and SST-2 datasets obtained via the Hugging Face `datasets` library. Text data was converted into numerical representations using TF-IDF vectorization, and a Logistic Regression model was trained for sentiment classification. This experiment demonstrated the effectiveness of classical feature engineering approaches for text classification tasks.

Week 2: Word Embeddings and LSTM Models

Week two introduced distributed word representations. Word2Vec embeddings were trained on a small corpus to capture semantic relationships between words. Pretrained GloVe embeddings were also explored for comparison.

An LSTM-based sentiment classification model was implemented using the IMDb dataset. Text sequences were tokenized, padded, and fed into an embedding layer followed by an LSTM network. The model demonstrated improved contextual understanding compared to TF-IDF based methods. Word embeddings were visualized using PCA and t-SNE to observe semantic clustering in lower-dimensional space.

Week 3: Transformer Models and Tokenization

In week three, transformer-based models were explored. A pre-trained BERT model was fine-tuned on the SST-2 dataset for sentiment classification using the Hugging Face `Trainer` API. This enabled efficient training and evaluation with minimal code.

A custom subword tokenizer was trained using the SentencePiece library, highlighting the importance of subword tokenization for handling rare and unseen words. Transformer-based inference was performed using the Hugging Face `pipeline` API, demonstrating how fine-tuned models can be deployed for real-world text classification tasks.

Week 4: Introduction to Speech Processing

The fourth week focused on speech processing fundamentals. Speech was studied as a time-domain digital signal with varying frequency content over time. Core audio concepts such as sampling rate, amplitude, bit depth, and the Nyquist theorem were explored.

Time–frequency representations including the Short-Time Fourier Transform (STFT), spectrograms, Mel spectrograms, and Mel Frequency Cepstral Coefficients (MFCCs) were implemented using the Librosa library. An end-to-end audio preprocessing pipeline was built involving audio loading, normalization, framing, feature extraction, and visualization.

Classical speech recognition approaches such as GMM-HMM systems were compared with modern neural and end-to-end models. The concept of Connectionist Temporal Classification (CTC) loss was studied to understand alignment-free training in speech recognition. A simple keyword spotting model was implemented using Mel spectrogram or MFCC features and a convolutional neural network on a subset of the Google Speech Commands dataset.

Conclusion

Across four weeks, this laboratory series provided practical exposure to both text and speech processing pipelines. The experiments demonstrated the evolution from classical feature-based methods to deep learning and transformer-based models, forming a strong foundation for advanced research and applications in NLP and Speech Processing.